

Quiz Result

B2 ADS Assignment

Subject: ADS

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Date: 9/21/2026

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Performance Summary

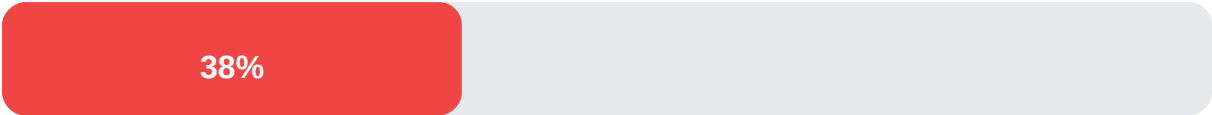
Your Score: 15 / 40

Percentage: 38%

Correct Answers: 10

Wrong Answers: 10

Questions Attempted: 20



38%

Score Breakdown

Raw Score (correct answers): +20

Total Penalty (wrong answers): -5

Final Score: 20 - 5 = 15

Difficulty Breakdown

Easy Questions Attempted: 0

Hard Questions Attempted: 20

Question-wise Breakdown

#	Question	Your Answer	Status	Score
1	Processing and analyzing extremely large datasets (petabytes or more) often requires distributed computing frameworks primarily due to which of the following challenges?	D. Memory and processing limitations of single machines	Correct	+2
2	A marketing firm wants to determine if the observed distribution of customer preferences for four new product designs (Design 1, Design 2, Design 3, Design 4) significantly deviates from an expected uniform distribution, where each design is expected to be equally preferred. They survey 200 customers. Which statistical test is most appropriate for this analysis?	C. A Chi-Square Goodness-of-Fit Test, to compare observed preferences to the expected uniform distribution.	Correct	+2
3	When communicating data insights, what is the most crucial difference in approach between a technical and a non-technical audience?	D. Technical audiences prefer more colorful visualizations, while non-technical audiences prefer simpler ones.	Wrong	-0.5
4	Which principle of Responsible AI specifically addresses the potential for AI systems to perpetuate or amplify societal biases present in training data?	B. Fairness	Correct	+2
5	A visualization showing average income by race and gender, where a specific racial group appears to have significantly lower income due to an underrepresentation of high-income individuals in the sample, but this bias is not disclosed, represents a failure in:	D. Transparency and fairness in visualization	Correct	+2
6	A model trained to predict house prices performs poorly because the relationship between features (e.g., number of rooms) and the target variable (price) has changed over time due to economic shifts. This scenario is an example of:	C. Concept Drift	Correct	+2
7	Which of the following is a common practice to avoid misleading visualizations?	C. Using inconsistent color palettes across different charts for the same metric.	Wrong	-0.5
8	How do SHAP and LIME primarily differ in their approach to generating model explanations?	A. SHAP provides global explanations, while LIME provides local explanations.	Wrong	-0.5
9	Which method is generally more robust for detecting outliers in a univariate dataset if the data distribution is heavily skewed?	B. Z-score method	Wrong	-0.5
10	What is a key principle of 'Fairness' in data science ethics?	C. Treating all data subjects equally and avoiding biased outcomes in models or visualizations.	Correct	+2
11	If continuous monitoring detects significant concept drift impacting a deployed model's performance, what is typically the most appropriate next step in an MLOps pipeline?	C. Trigger a full model retraining process with updated data and potentially new features/models	Correct	+2
12	A company wants to develop a personalized recommendation system for its e-commerce platform. Which data science role would be most critically involved in designing, building, and optimizing the predictive algorithms for this	A. Machine Learning Engineer / Data Scientist	Correct	+2

#	Question	Your Answer	Status	Score
	system?			
13	When deploying an ML model for real-time inference, what is the primary advantage of exposing it via a REST API using frameworks like Flask or FastAPI?	C. It guarantees continuous model retraining.	Wrong	-0.5
14	In the banking sector, what is a common application of machine learning that helps financial institutions mitigate financial loss by identifying unusual transactions?	B. Fraud detection	Correct	+2
15	A sports scientist wants to evaluate the effectiveness of a new training program. They recruit 30 athletes and measure their vertical jump height *before* starting the program. After 8 weeks of the program, they measure the *same 30 athletes'* vertical jump height *again*. To determine if the training program significantly improved vertical jump height, which statistical test is most appropriate?	A. An independent samples t-test, comparing the mean jump height of the 'before' group to the 'after' group.	Wrong	-0.5
16	A researcher wants to compare the average reaction times of participants under four different experimental conditions (A, B, C, D). They could perform multiple independent samples t-tests (e.g., A vs B, A vs C, A vs D, B vs C, B vs D, C vs D). Alternatively, they could perform a One-Way ANOVA. What is the primary statistical advantage of using a One-Way ANOVA over multiple t-tests in this situation?	D. ANOVA is computationally simpler and faster for more than two groups.	Wrong	-0.5
17	A study investigates the association between preferred mode of transport (Car, Bus, Train, Bicycle) and residential area (Urban, Suburban, Rural). The data is collected from 100 participants. After constructing a contingency table, the researcher finds that 2 cells have an expected count of less than 1, and 5 cells have an expected count between 1 and 5. Which of the following is the most appropriate course of action?	C. A Chi-Square Test is not suitable; a non-parametric test for ordinal data should be considered if the categories can be ordered.	Wrong	-0.5
18	A social scientist performs a Chi-Square test for independence on a 3x4 contingency table (3 rows, 4 columns) with a total sample size of N=500. The test yields a p-value of 0.008. Which of the following is the most accurate interpretation of this result in isolation?	D. The sample size of 500 is insufficient to reliably detect an association in a 3x4 table, necessitating a larger sample or an exact test.	Wrong	-0.5
19	A distribution described as 'leptokurtic' has:	C. Tails and peak similar to a normal distribution	Wrong	-0.5
20	What is the primary purpose of data normalization in the data preprocessing phase?	A. To scale numerical features to a standard range	Correct	+2

Correct Answers for Wrong Responses

Q#	Question	Correct Answer
3	When communicating data insights, what is the most crucial difference in approach between a technical and a non-technical audience?	C. Technical audiences require deep dives into methodology and statistical significance, whereas non-technical audiences need clear, actionable takeaways.

Q#	Question	Correct Answer
7	Which of the following is a common practice to avoid misleading visualizations?	D. Always starting the y-axis at zero for bar charts showing absolute values.
8	How do SHAP and LIME primarily differ in their approach to generating model explanations?	D. SHAP is based on game theory for feature contributions, while LIME creates local surrogate models.
9	Which method is generally more robust for detecting outliers in a univariate dataset if the data distribution is heavily skewed?	C. Interquartile Range (IQR) method
13	When deploying an ML model for real-time inference, what is the primary advantage of exposing it via a REST API using frameworks like Flask or FastAPI?	A. It provides a standardized, language-agnostic way for other applications to request predictions.
15	A sports scientist wants to evaluate the effectiveness of a new training program. They recruit 30 athletes and measure their vertical jump height *before* starting the program. After 8 weeks of the program, they measure the *same 30 athletes* vertical jump height *again*. To determine if the training program significantly improved vertical jump height, which statistical test is most appropriate?	D. A paired samples t-test, comparing the 'before' and 'after' measurements for each athlete.
16	A researcher wants to compare the average reaction times of participants under four different experimental conditions (A, B, C, D). They could perform multiple independent samples t-tests (e.g., A vs B, A vs C, A vs D, B vs C, B vs D, C vs D). Alternatively, they could perform a One-Way ANOVA. What is the primary statistical advantage of using a One-Way ANOVA over multiple t-tests in this situation?	B. ANOVA reduces the Type I error rate (false positive) for the family of comparisons.
17	A study investigates the association between preferred mode of transport (Car, Bus, Train, Bicycle) and residential area (Urban, Suburban, Rural). The data is collected from 100 participants. After constructing a contingency table, the researcher finds that 2 cells have an expected count of less than 1, and 5 cells have an expected count between 1 and 5. Which of the following is the most appropriate course of action?	A. Combine categories with low expected counts (e.g., merge 'Train' and 'Bicycle' or 'Rural' and 'Suburban') to meet the assumption, then re-run the Chi-Square test.
18	A social scientist performs a Chi-Square test for independence on a 3x4 contingency table (3 rows, 4 columns) with a total sample size of N=500. The test yields a p-value of 0.008. Which of the following is the most accurate interpretation of this result in isolation?	B. The observed cell frequencies across the entire table are unlikely to have occurred by chance if the variables were truly independent.
19	A distribution described as 'leptokurtic' has:	A. Thicker tails and a higher peak than a normal distribution